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Sadlerian Professor of Pure Mathematics in the University of Cambridge.
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Advertisement (book p. v).

The present volume contains fifty-eight papers (numbered 101, 102, . . . , 158) originally published, all but two of them, in the years 1851 to 1860: they are here reproduced nearly but not exactly in chronological order. The two excepted papers are 142, *Numerical Tables Supplementary to Second Memoir on Quantics*, now first published (1889); and, 143, *Tables of the Covariants M to W of the Binary Quintic*: from the Second, Third, Fourth, Fifth, Eighth, Ninth and Tenth Memoirs on Quantics (arranged in the present form, 1889): the determination of the finite number, 23, of the covariants of the quintic was made by Gordan in the year 1869, and the calculation of them having been completed in my Ninth and Tenth Memoirs, it appeared to me convenient in the present republication to unite together the values of all the covariants: viz. those of A to L are given in the Second Memoir 141, and the remainder M to W in the paper 143. I have added to the Third Memoir 144, in the notation thereof, some formulae which on account of a difference of notation were omitted from a former paper, 35.

I remark that the present volume comprises the first six of the ten Memoirs on Quantics, viz. these are 139, 141, 144, 155, 156 and 158. I have, in the Notes and References, inserted a discussion of some length in reference to the paper 121, *Note on a Question in the Theory of Probabilities*: and also some remarks in reference to the theory of Distance developed in the Sixth Memoir on Quantics, 158.

A. C.

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Contents (book pp. vii–xi).

	PAGE
101. <i>Notes on Lagrange's Theorem</i>	1
Camb. and Dubl. Math. Jour. t. vi. (1851), pp. 37—45	
102. <i>On a Double Infinite Series</i>	8
Camb. and Dubl. Math. Jour. t. vi. (1851), pp. 45—47	
103. <i>On Certain Definite Integrals</i>	11
Camb. and Dubl. Math. Jour. t. vi. (1851), pp. 136—140	
104. <i>On the Theory of Permutants</i>	16
Camb. and Dubl. Math. Jour. t. vii. (1852), pp. 40—51	
105. <i>Correction to the Postscript to the Paper on Permutants</i>	27
Camb. and Dubl. Math. Jour. t. vii. (1852), pp. 97—98	
106. <i>On the Singularities of Surfaces</i>	28
Camb. and Dubl. Math. Jour. t. vii. (1852), pp. 166—171	
107. <i>On the Theory of Skew Surfaces</i>	33
Camb. and Dubl. Math. Jour. t. vii. (1852), pp. 171—173	
108. <i>On certain Multiple Integrals connected with the Theory of Attractions</i>	35
Camb. and Dubl. Math. Jour. t. vii. (1852), pp. 174—178	
109. <i>On the Rationalisation of certain Algebraical Equations</i>	40
Camb. and Dubl. Math. Jour. t. viii. (1853), pp. 97—101	
110. <i>Note on the Transformation of a Trigonometrical Expression</i>	45
Camb. and Dubl. Math. Jour. t. ix. (1854), pp. 61—62	
111. <i>On a Theorem of M. Lejeune-Dirichlet's</i>	47
Camb. and Dubl. Math. Jour. t. ix. (1854), pp. 163—165	
112. <i>Demonstration of a Theorem relating to the Products of Sums of Squares</i>	49
Phil. Mag. t. iv. (1852), pp. 515—519	
113. <i>On the Geometrical Representation of the Integral $\int dx \div \sqrt{(x+a)(x+b)(x+c)}$</i>	53
Phil. Mag. t. v. (1853), pp. 281—284	
114. <i>Analytical Researches connected with Steiner's Extension of Malfatti's Problem</i> ..	57
Phil. Trans. t. cxlii. (for 1852), pp. 253—278	
115. <i>Note on the Porism of the In-and-circumscribed Polygon</i>	87
Phil. Mag. t. vi. (1853), pp. 99—102	
116. <i>Correction of two Theorems relating to the In-and-circumscribed Polygon</i>	91
Phil. Mag. t. vi. (1853), pp. 376—377	
117. <i>Note on the Integral $\int dx \div \sqrt{(m-x)(x+a)(x+b)(x+c)}$</i>	93
Phil. Mag. t. vi. (1853), pp. 103—105	
118. <i>On the Harmonic Relation of two Lines or two Points</i>	96
Phil. Mag. t. vi. (1853), pp. 105—107	
119. <i>On a Theorem for the Development of a Factorial</i>	98
Phil. Mag. t. vi. (1853), pp. 182—185	
120. <i>Note on a Generalisation of the Binomial Theorem</i>	102
Phil. Mag. t. vi. (1853), p. 185	
121. <i>Note on a Question in the Theory of Probabilities</i>	103
Phil. Mag. t. vi. (1853), p. 259	
122. <i>On the Homographic Transformation of a Surface of the Second Order into Itself</i> ..	105
Phil. Mag. t. vi. (1853), pp. 326—333	
123. <i>On the Geometrical Representation of an Abelian Integral</i>	113
Phil. Mag. t. vi. (1853), pp. 414—418	
124. <i>On a Property of the Caustic by Refraction of the Circle</i>	118
Phil. Mag. t. vi. (1853), pp. 427—431	

	PAGE
125. <i>On the Theory of Groups as depending on the Symbolical Equation $\theta^n = 1$</i>	123
Phil. Mag. t. vii. (1854), pp. 40—47	
126. <i>On the theory of Groups as depending on the Symbolical Equation $\theta^n = 1$. Second Part</i>	131
Phil. Mag. t. vii. (1854), pp. 408—409	
127. <i>On the Homographic Transformation of a Surface of the Second Order into itself</i>	133
Phil. Mag. t. vii. (1854), pp. 208—212: continuation of 122	
128. <i>Developments on the Porism of the In-and-circumscribed Polygon</i>	138
Phil. Mag. t. vii. (1854), pp. 339—345	
129. <i>On the Porism of the In-and-circumscribed Triangle, and on an irrational Transformation of two Ternary Quadratic Forms each into itself</i>	145
Phil. Mag. t. ix. (1855), pp. 513—517	
130. <i>Deuxième Mémoire sur les Fonctions doublement Périodiques</i>	150
Liouville, t. xix. (1854), pp. 193—208: sequel to 25	
131. <i>Nouvelles Recherches sur les Covariants</i>	164
Crelle, t. xlvii. (1854), pp. 109—125	
132. <i>Réponse à une Question proposée par M. Steiner</i>	179
Crelle, t. l. (1855), pp. 277—278	
133. <i>Sur un Théorème de M. Schläfli</i>	181
Crelle, t. l. (1855), pp. 278—282	
134. <i>Remarques sur la Notation des Fonctions Algébriques</i>	185
Crelle, t. l. (1855), pp. 282—285	
135. <i>Note sur les Covariants d'une Fonction Quadratique, Cubique, ou Biquadratique à deux Indéterminées</i>	189
Crelle, t. l. (1855), pp. 285—287	
136. <i>Sur la Transformation d'une Fonction Quadratique en elle-même par des Substitutions linéaires</i>	192
Crelle, t. l. (1855), pp. 288—299	
137. <i>Recherches Ultérieures sur les Déterminants gauches</i>	202
Crelle, t. l. (1855), pp. 299—313: continuation of 52 and 69	
138. <i>Recherches sur les Matrices dont les termes sont des fonctions linéaires d'une seule Indéterminée</i>	216
Crelle, t. l. (1855), pp. 313—317	
139. <i>An Introductory Memoir on Quantics</i>	221
Phil. Trans. t. cxliv. (for 1854), pp. 244—258	
140. <i>Researches on the Partition of Numbers</i>	235
Phil. Trans. t. cxlv. (for 1855), pp. 127—140	
141. <i>A Second Memoir on Quantics</i>	250
Phil. Trans. t. cxlvi. (for 1856), pp. 101—126	
142. <i>Numerical Tables Supplementary to Second Memoir on Quantics</i>	276
Now first published (1889)	
143. <i>Tables of the Covariants M to W of the Binary Quintic: from the Second, Third, Fifth, Eighth, Ninth and Tenth Memoirs on Quantics</i>	282
Arranged in the present form, 1889	
144. <i>A Third Memoir on Quantics</i>	310
Phil. Trans. t. cxlvi. (for 1856), pp. 627—647	
145. <i>A Memoir on Caustics</i>	336
Phil. Trans. t. cxlvii. (for 1857), pp. 273—312	
146. <i>A Memoir on Curves of the Third Order</i>	381
Phil. Trans. t. cxlvii. (for 1857), pp. 415—446	
147. <i>A Memoir on the Symmetric Functions of the Roots of an Equation</i>	417
Phil. Trans. t. cxlvii. (for 1857), pp. 489—496	

	PAGE
148. <i>A Memoir on the Resultant of a System of two Equations</i>	440
Phil. Trans. t. cxlvii. (for 1857), pp. 703—715	
149. <i>On the Symmetric Functions of the Roots of certain Systems of two Equations</i> ...	454
Phil. Trans. t. cxlvii. (for 1857), pp. 717—726	
150. <i>A Memoir on the Conditions for the Existence of given Systems of Equalities among the Roots of an Equation</i>	465
Phil. Trans. t. cxlvii. (for 1857), pp. 727—731	
151. <i>Tables of the Sturmian Functions for Equations of the Second, Third, Fourth, and Fifth Degrees</i>	471
Phil. Trans. t. cxlvii. (for 1857), pp. 733—736	
152. <i>A Memoir on the Theory of Matrices</i>	475
Phil. Trans. t. cxlviii. (for 1858), pp. 17—37	
153. <i>A Memoir on the Automorphic Linear Transformation of a Bipartite Quadric Function</i>	497
Phil. Trans. t. cxlviii. (for 1858), pp. 39—46	
154. <i>Supplementary Researches on the Partition of Numbers</i>	506
Phil. Trans. t. cxlviii. (for 1858), pp. 47—52	
155. <i>A Fourth Memoir on Quantics</i>	513
Phil. Trans. t. cxlviii. (for 1858), pp. 415—427	
156. <i>A Fifth Memoir on Quantics</i>	527
Phil. Trans. t. cxlviii. (for 1858), pp. 429—460	
157. <i>On the Tangential of a Cubic</i>	558
Phil. Trans. t. cxlviii. (for 1858), pp. 461—463	
158. <i>A Sixth Memoir on Quantics</i>	561
Phil. Trans. t. cxlix. (for 1859), pp. 61—90	
<i>Notes and References</i>	593

Classification (book p. xii).

Geometry.

- Theory of Distance, 158
- Surfaces, 106, 107
- Transformation of Quadric Surfaces, 122, 127, 129, 136, 153
- Steiner's extension of Malfatti's Problem, 114
- In-and-circumscribed triangle and polygon, 115, 116, 128, 129
- Harmonic relation of two lines or points, 118
- Question proposed by Steiner, 132
- Caustics, 124, 145
- Cubic Curves, 146, 157

Analysis.

- Skew Determinants, 137
- Attractions and Multiple Integrals, 108
- Definite Integrals, 103
- Elliptic and Abelian Integrals, 110, 113, 117, 123, 130
- Covariants, Quantics &c., 131, 134, 135, 139, 141, 142, 143, 144, 155, 156, 158
- Matrices, 138, 152
- Partition of Numbers, 140, 154
- Symmetric Functions &c., 147, 148, 149, 150
- Lagrange's Theorem, 101
- Double Infinite Series, 102
- Permutants, 104, 105
- Rationalisation of Algebraic Expression, 109
- Transformation of Trigonometrical Expression, 110
- Theorem of Lejeune-Dirichlet's, 111
- Products of Sums of Squares, 112
- Factorials, 119
- Generalisation of Binomial Theorem, 120
- Question in Probabilities, 121
- Groups, 125, 126
- Theorem of Schläfli's, on Elimination, 133
- Sturmian Functions, 151

101. Notes on Lagrange's Theorem.

[From the *Cambridge and Dublin Mathematical Journal*, vol. vi. (1851), pp. 37—45.]

I.

If in the ordinary form of Lagrange's theorem we write $(x + a)$ for x , it becomes

$$\begin{aligned} x &= hf(a + x), \\ F(a + x) &= Fa + \frac{h}{1} F'a fa + \&c. \end{aligned} \quad (1)$$

It follows that the equation

$$F(a + x) = Fa + \sum \frac{1}{r} \frac{x^{r-1}}{[r-1]} \frac{d^{r-1}}{da^{r-1}} (F'a f^r a) + \dots \quad (2)$$

must reduce itself to an identity when the two sides are expanded in powers of x ; or writing for shortness F , f instead of Fa , fa , and δ for $\frac{d}{da}$, we must have

$$\frac{1}{[r]^r} \delta^{r-1} F = \delta \sum \frac{1}{[p-1]^{p-1} [r-p]^{r-p}} \delta^{p-1} (\delta F \cdot f^p) \cdot \delta^{r-p} f^{-p} \cdot \frac{1}{p}, \quad (3)$$

(where p extends from 1 to r). Or what comes to the same,

$$\frac{1}{[r]^r} \delta^{r-1} F = \sum \frac{1}{[s]^s [p] [r-p]^{r-p}} \delta^{s-1} \{ (\delta F \cdot f^p) \delta^{r-p} f^{-p} \},$$

where s extends from 0 to $(r-p)$. The terms on the two sides which involve $\delta^r F$ are immediately seen to be equal; the coefficients of the remaining terms $\delta^s F$ on the second side must vanish, or we must have

$$\sum \frac{1}{[s]^s} \frac{1}{[p-1]^{p-1} [r-p]^{r-p}} \delta^{r-s} \{ f^p \} \delta^{r-p} f^{-p} = 0 \quad (4)$$

(s being less than r). Or in a somewhat more convenient form, writing p , q and k for $p-s$, $r-p$ and $r-s$,

$$\sum \frac{1}{[p]^p [q]^q} \delta^p f^{p+s} \delta^q f^{-(p+s)} = 0, \quad (5)$$

where s is constant and p and q vary subject to $p+q=k$, k being a given constant different from zero (in the case where $k=0$, the series reduces itself to the single term $\frac{1}{s}$). The direct proof of this theorem will be given presently.

II.

The following symbolical form of Lagrange's theorem was given by me in the *Mathematical Journal*, vol. iii. [1843], pp. 283—286, [8].

If

$$x = a + hfx, \quad (6)$$

then

$$Fx = \left(\frac{d}{da} \right)^{-1} \frac{d}{da} Fa \cdot e^{hfa}.$$

Suppose $fx = \phi(b + k\psi x)$, or $x = a + h\phi(b + k\psi x)$, then

$$Fx = \left(\frac{d}{da} \right)^{-1} \frac{d}{da} Fa \cdot e^{h\phi(b+k\psi a)}.$$

But

$$\phi(b + k\psi a) = \left(\frac{d}{db} \right)^{-1} \frac{d}{db} \phi b \cdot e^{k\psi a}.$$

(In fact the two general terms

$$\{\phi(b + k\psi a)\}^m \quad \text{and} \quad \left(\frac{d}{db}\right)^{-1} e^{k\psi a} (\phi b)^m,$$

of which the former reduces itself to $e^{k\psi a} \cdot (\phi b)^m$, are equal on account of the equivalence of the symbols

$$e^{k\psi a} \quad \text{and} \quad \left(\frac{d}{db}\right)^{-1} \frac{d}{db} e^{k\psi a}.$$

Hence

$$\begin{aligned} x &= a + h\phi(b + k\psi x), \\ Fx &= \left(\frac{d}{da}\right)^{-1} \frac{d}{da} \left(\frac{d}{db}\right)^{-1} \frac{d}{db} Fa \cdot e^{h\phi b + k\psi a}, \end{aligned} \tag{7}$$

and the coefficient of $h^m k^n$ is

$$\frac{1}{[m]^m [n]^n} \left(\frac{d}{da}\right)^{m-1} \left(\frac{d}{db}\right)^{n-1} F'a (\psi a)^n (\phi b)^m.$$

A similar formula evidently applies to the case of any finite number of functions $\phi, \psi, \&c.$: in the case of an infinite number we have

$$F(a + h\phi(b + k\psi(c + l\chi(d + \dots))) = \left(\frac{d}{da}\right)^{-1} \left(\frac{d}{db}\right)^{-1} \left(\frac{d}{dc}\right)^{-1} \dots F'a e^{h\phi b + k\psi c + l\chi d + \dots}; \tag{8}$$

or the coefficient of $h^m k^n l^p \dots$ is

$$\frac{1}{[m]^m [n]^n [p]^p \dots} \left(\frac{d}{da}\right)^{m-1} \left(\frac{d}{db}\right)^{n-1} \left(\frac{d}{dc}\right)^{p-1} \dots F'a (\phi b)^m (\psi c)^n (\chi d)^p \dots,$$

the last of the series $m, n, p \dots$ being always zero; e.g. in the coefficient of $h^m k^n$, account must be had of the factor $\left(\frac{d}{dc}\right)^{-1} (\chi c)^0$ or $(\chi c)^0$. The above form is readily proved independently by Taylor's theorem, without the assistance of Lagrange's. If in it we write $h = k = \&c.$, $a = b = \&c.$, and $\phi = \psi = \&c. = f$, we have $F(a + hf(a + hf(a + \dots))) = Fx$, where $x = a + hfx$. Hence, comparing the coefficient of h^s with that given by Lagrange's theorem,

$$\frac{1}{[s]^s} \delta^{s-1} (\delta F \cdot f^s) = \sum \frac{1}{[m]^m [n]^n \dots} \delta^{m-1} \delta F \cdot f^m \cdot \delta^{n-1} f^n \dots, \tag{9}$$

where $m + n + \&c. = s$, and as before $Fa, fa, \frac{d}{da}$ have been replaced by F, f, δ . By comparing the coefficients of $h^s F$,

$$\frac{1}{[t]^t} \delta^t f^t = \sum \frac{1}{[n]^n [p]^p \dots} \delta^{n-1} f^n \cdot \delta^{p-1} f^p \dots, \tag{10}$$

where $n + p + \dots = t$, the last of the series $n, p \dots$ always vanishing. The formula (10) deduced, as above mentioned, from Taylor's theorem, and the subsequent formula (11) with an independent demonstration of it, not I believe materially different from that which will presently be given, are to be found in a memoir by M. Collins (volume ii. (1833) of the Memoirs of the Academy of St Petersburg), who appears to have made very extensive researches in the theory of developments as connected with the combinatorial analysis.

III.

To demonstrate the formula (6), consider, in the first place, the expression

$$\sum \frac{\phi p}{[p]^p [q]^q} (\delta^p f^{p+s}) (\delta^q f^{-p-s}),$$

where $p + q = k$. Since

$$\frac{1}{[p]^p [q]^q} = \frac{1}{k} \left(\frac{1}{[p-1]^{p-1} [q]^q} + \frac{1}{[p]^p [q-1]^{q-1}} \right),$$

this is immediately transformed into

$$\frac{1}{k} \sum \delta \phi p \cdot \frac{1}{[p-1]^{p-1} [q]^q} (\delta^{p-1} f^{p+s}) (\delta^q f^{-p-s}),$$

in which last expression $p + q = (p - 1)$. Of this, after separating the factor δf , the general term is

$$\frac{1}{k} \cdot \frac{1}{[p]^p [q]^q} \{ \phi(p + s + 1) (\delta^p f^{p+s+1}) (\delta^q f^{-p-s-1}) \},$$

equivalent to

$$\begin{aligned} \frac{1}{k} \cdot \frac{1}{[p]^p [q]^q} \delta^{s+1} f \cdot \sum \frac{1}{[p]^p [q]^q} \{ \phi(p + a + 1) (p + s + a + 1) (\delta^p f^{p+s+a}) (\delta^q f^{-p-s-a}) \\ - \phi(p + s + a) (\delta^p f^{p+a}) (\delta^q f^{-p-a-s}) \}, \end{aligned}$$

in which last expression $p + q = k - a - 1$. By repeating the reduction j times, the general term becomes

$$\begin{aligned} \frac{1}{k(k - \alpha - 1)(k - \alpha - \beta - 2) \dots} \frac{1}{[\alpha]^\alpha [\beta]^\beta \dots} \delta^{s+\alpha} f \delta^{s+\beta} f \dots \\ \times \sum \frac{1}{[p]^p [q]^q} \Sigma \{ (-)^{j-f} \phi(p + \alpha + \beta \dots + f) (p + s + \alpha + \beta \dots + f) \\ \times [p + s + \alpha + \beta \dots + j - 1]^{j-f} (\delta^p f^{p+s+\alpha+\dots+f}) (\delta^q f^{-p-s-\alpha-\dots-f}) \}, \end{aligned}$$

where the sums $\alpha + \beta \dots$ contain f terms, f being less than j or equal to it, and Σ extends to all combinations of the quantities $\alpha, \beta \dots$ taken f and f together (so that the summation contains 2^j terms). Also $p + q = k - \alpha - \beta \dots (j \text{ terms}) - j$, and the products $k(k - \alpha - 1)(k - \alpha - \beta - 2) \dots$ and $[\alpha]^\alpha [\beta]^\beta \dots \delta^{s+\alpha} f \delta^{s+\beta} f \dots$ contain each of them j terms. Suppose the reduction continued until $k - \alpha - \beta \dots (j \text{ terms}) - j = 0$, then the only values of p, q are $p = 0, q = 0$; and the general term of

$$\sum \frac{\phi p}{[p]^p [q]^q} (\delta^p f^{p+s}) (\delta^q f^{-p-s})$$

becomes

$$\begin{aligned} \frac{1}{k(k - \alpha - 1)(k - \alpha - \beta - 2) \dots} \frac{1}{[\alpha]^\alpha [\beta]^\beta \dots} \delta^{s+\alpha} f \delta^{s+\beta} f \dots f^s \cdot f^{-s} \\ \times \Sigma \{ (-)^{j-f} \phi(\alpha + \beta \dots + f) [s + \alpha + \beta \dots + f]^f [s + \alpha + \beta \dots + j - 1]^{j-f} \}. \end{aligned}$$

If $\phi p = \frac{1}{p+s}$, the general term reduces itself to

$$\begin{aligned} \frac{1}{k(k - \alpha - 1)(k - \alpha - \beta - 2) \dots} \frac{1}{[\alpha]^\alpha [\beta]^\beta \dots} \delta^{s+\alpha} f \delta^{s+\beta} f \dots f^s \cdot f^{-s} \\ \sum \{ (-)^{j-f} (s + \alpha + \beta \dots + f) \phi(\alpha + \beta \dots + f) [s + \alpha + \beta \dots + j - 1]^{j-f} \}; \end{aligned}$$

whence finally, if $\phi p = \frac{1}{p+s}$, the general term of

$$\sum \frac{\phi p}{[p]^p [q]^q} (\delta^p f^{p+s}) (\delta^q f^{-p-s})$$

becomes

$$\begin{aligned} \frac{1}{k(k - \alpha - 1)(k - \alpha - \beta - 2) \dots} \frac{1}{[\alpha]^\alpha [\beta]^\beta \dots} \delta^{s+\alpha} f \delta^{s+\beta} f \dots f^s \cdot f^{-s} \\ \times \sum \{ (-)^{j-f} \phi(\alpha + \beta \dots + f) [s + \alpha + \beta \dots + j - 1]^{j-f} \}, \end{aligned}$$

and it is readily shown that the sum contained in this formula vanishes, which proves the equation in question.

IV.

The demonstration of the equation (11) is much simpler. We have

$$\delta^t(f^{-t} \cdot \delta f) = \sum \binom{t-1}{n-1} \delta^n(f^{-t} \cdot \delta f)(\delta^{t-n} f^n),$$

that is,

$$\delta^t f^t = \frac{s}{s-t} \sum \binom{t-1}{n-1} (\delta^n f^{-t})(\delta^{t-n} f^n),$$

where n extends from $n = 1$ to $n = t$. Similarly

$$\delta^{t-n} f^n = \frac{t-n}{p} \sum \binom{t-n-p-1}{p-1} (\delta^p f^n)(\delta^{t-n-p} f^{n-p}),$$

&c.

Hence, substituting successively, and putting $t - n - p - q = r$, &c.,

$$\delta^t f^t = \frac{s}{s-t} \sum \frac{[t]^t}{[n]^n [p]^p (q+r) [q-1]^{q-1} \dots} (\delta^n f^n)(\delta^p f^p)(\delta^q f^q) \dots,$$

&c.; and the last of these corresponding to a zero value of the last of the quantities $n, p, q \dots$ is evidently the required equation (11).

V.

The formula (18) in my paper on Lagrange's theorem (before referred to) is incorrect. I propose at present, after giving the proper form of the formula in question, to develop the result of the substitution indicated at the conclusion of the paper. It will be convenient to call to mind the general theorem, that when any number of variables $x, y, z \dots$ are connected with as many other variables $u, v, w \dots$ by the same number of equations (so that the variables of each set may be considered as functions of those of the other set) the quotient of the expressions $dx dy \dots$ and $du dv \dots$ is equal to the quotient of two determinants formed with the functions which equated to zero express the relations between the two sets of variables; the former with the differential coefficients of these functions with respect to $u, v \dots$, the latter with the differential coefficients with respect to $x, y \dots$. Consequently the notation $\frac{dx dy \dots}{du dv \dots}$ may be considered as representing the quotient of these determinants. This being premised, if we write

$$x - u - h\theta(x, y \dots) = 0,$$

$$y - v - k\phi(x, y \dots) = 0,$$

then the formula in question is

$$F(x, y \dots) = \delta_u^{-1} \delta_v^{-1} \dots e^{h\theta + k\phi + \dots} F \nabla,$$

when for shortness $u, v, \dots$ are the letters considered as a function of $\theta, \phi, \dots$ substituted for $x, y \dots$. Let $\frac{X}{V}$ denote the value which $\frac{dx dy \dots}{du dv \dots}$ assumes when these variables are changed into $u, v, \dots$, we have

$$\nabla = \begin{vmatrix} 1 - h\delta_u \theta & -h\delta_v \theta & \dots \\ -k\delta_u \phi & 1 - k\delta_v \phi & \dots \\ \vdots & & \end{vmatrix}.$$

By changing the function F , we obtain

$$F(x, y \dots) = \delta_u^{m-1} \delta_v^{n-1} \dots e^{h\theta + k\phi + \dots} F \nabla,$$

where, however, it must be remembered that the h, k in so far as they enter into the function ∇ , are not affected by the symbols $h\delta_u, k\delta_v, \dots$. In order that we may consider them to be so affected, it is necessary

in the function ∇ to replace $h, k, \&c.$ by $\frac{h}{\delta_u}, \frac{k}{\delta_v}, \&c.$ When this is done, observing that the symbols $h\delta_u\theta, k\delta_v\theta \dots$ affect a function $e^{h\theta+k\phi+\dots}$, the symbols $h\delta_u\theta, k\delta_v\theta, \dots$ may be replaced by δ_u^*, δ_v^* , where the $*$ is not an index, but an affix denoting that the differentiation is only to be performed with respect to $u, v \dots$ so far as these variables respectively enter into the function θ . Transforming the other lines of the determinant in the same manner, and taking out from $\delta_u^m \delta_v^n \dots$ the factor $\delta_u \delta_v \dots$ in order to multiply this last factor into the determinant, we obtain

$$F(x, y \dots) = \delta_u^{m-1} \delta_v^{n-1} \dots e^{h\theta+k\phi+\dots} F\Box;$$

where

$$\Box = \begin{vmatrix} \delta_u - \delta_u^\theta & -\delta_v^\theta & \dots \\ -\delta_u^\phi & \delta_v - \delta_v^\phi & \dots \\ \vdots & & \end{vmatrix},$$

in which expression $\delta_u, \delta_v, \dots$ are to be replaced by

$$\delta_u + \delta_u^\theta + \delta_u^\phi + \delta_u^\chi + \dots.$$

The complete expansion is easily arrived at by induction, and the form is somewhat singular. In the case of a single variable u we have $\Box = \delta_u^\theta$; in the case of two variables, $\Box = \delta_u^\theta \delta_v^\phi + \delta_u^\phi \delta_v^\theta + \delta_u^\theta \delta_v^\theta$. Or writing down only the affixes, in the case of a single variable we have F ; in the case of two variables $FF, F\theta, \phi F$; and in the case of three variables $FFF, \phi FF, \chi \phi F, F\chi F, F\theta F, FF\theta, FF\phi, F\theta\theta, F\theta\phi, F\chi\theta, \phi F\phi, \chi F\phi, \phi F\theta, \chi\chi\phi, \phi\chi\theta, \chi\theta\theta$, where it will be observed that θ never occurs in the first place, nor ϕ in the second place, nor θ, ϕ (in any order) in the first and second places, $\&c.$, nor θ, ϕ, χ (in any order) in the first, second, and third places. And the same property holds in the general case for each letter and binary, ternary, $\&c.$ combination, and for the entire system of letters, and the system of affixes contains every possible combination of letters not excluded by the rule just given. Thus in the case of two letters, forming the system of affixes $FF, F\theta, \phi F, \theta F, F\phi, \theta\phi, \phi\theta$, the last four are excluded, the first three of them by containing θ in the first place or ϕ in the second place, the last by containing θ, ϕ in the first and second places: and there remains only the terms $FF, F\theta, \phi F$ forming the system given above. Substituting the expanded value of $\Box$ in the expression for $F(x, y \dots)$, the equation may either be permitted to remain in the form which it thus assumes, or we may, in order to obtain the finally reduced form, after expanding the powers of $h, k \dots$, connect the symbols $\delta_u^\theta, \delta_v^\phi \dots, \delta_u^\chi, \&c.$ with the corresponding functions $\theta, \phi \dots F$, and then omit the affixes; thus, in particular, in the case of a single variable the general term of Fx is

$$\frac{h^p}{[p]^p} \delta_u^{p-1} (F' \cdot \theta^p)$$

(the ordinary form of Lagrange's theorem). In the case of two letters the general term of $F(x, y)$ is

$$\frac{h^p k^q}{[p]^p [q]^q} \delta_u^{p-1} \delta_v^{q-1} [\theta^p \phi^q F' F' + p \theta^{p-1} \phi^q \delta_v \theta F' F + q \theta^p \phi^{q-1} \delta_u \phi F F']$$

(see the *Mécanique Céleste*, [Ed. 1, 1798] t. i. p. 176). In the case of three variables, the general term is

$$\frac{h^p k^q l^r}{[p]^p [q]^q [r]^r} \delta_u^{p-1} \delta_v^{q-1} \delta_w^{r-1} [\theta^p \phi^q \chi^r F' F' F' + \dots],$$

the sixteen terms within the $\{ \}$ being found by comparing the product $\delta_u \delta_v \delta_w$ with the system $FFF, \phi FF, \&c.$, given above, and then connecting each symbol of differentiation with the function corresponding to the affix. Thus in the first term the $\delta_u, \delta_v, \delta_w$ each affect the F , in the second term the δ_u affects ϕ^p , and the δ_v and δ_w each affect the F , and so on for the remaining terms. The form is of course deducible from Laplace's general theorem, and the actual development of it is given in Laplace's Memoir in the *Hist. de l'Acad.* vii. I quote from a memoir by Jacobi which I take this opportunity of referring to, "De resolutione equationum per series infinitas," Crelle, t. vi. [1830], pp. 257—286, founded on a preceding memoir, "Exercitatio algebraica circa discriptionem singularem fractionum quæ plures variables involvunt," t. v. [1830], pp. 344—364.

Stone Buildings, April 6, 1850.

102. On a Double Infinite Series.

[From the *Cambridge and Dublin Mathematical Journal*, vol. vi. (1851), pp. 45—47.]

The following completely paradoxical investigation of the properties of the function Γ (which I have been in possession of for some years) may perhaps be found interesting from its connexion with the theories of expansion and divergent series.

Let $\Sigma_r \phi r$ denote the sum of the values of ϕ for all integer values of r from $-\infty$ to ∞ . Then writing

$$u = \Sigma_r [n-1]^r x^{n-1-r}, \quad (11)$$

(where n is any number whatever), we have immediately

$$\frac{du}{dx} = \Sigma_r [n-1]^{r+1} x^{n-2-r} = \Sigma_r [n-1]^r x^{n-1-r} = u;$$

that is,

$$\frac{du}{dx} = u, \quad \text{or} \quad u = C_n e^x,$$

(the constant of integration being of course in general a function of n). Hence

$$C_n e^x = \Sigma_r [n-1]^r x^{n-1-r}; \quad (12)$$

or e^x is expanded in general in a doubly infinite necessarily divergent series of fractional powers of x , (which resolves itself however in the case of n a positive or negative integer, into the ordinary singly infinite series, the value of C_n in this case being immediately seen to be Γn).

The equation (2) in its general form is to be considered as a definition of the function C_n . We deduce from it

$$\Sigma_r [n-1]^r (ax)^{n-1-r} = C_n e^{ax},$$

and also

$$\Sigma_k [n+n' \dots -1]^k \{a(x+x' \dots)\}^{n+n' \dots -1-k} = C_{n+n' \dots} e^{a(x+x' \dots)}.$$

Multiplying the first set of series, and comparing with this last,

$$C_{n+n' \dots} \Sigma_{r,r' \dots} [n-1]^r [n'-1]^{r'} \dots x^{n-1-r} x'^{n'-1-r'} \dots = C_n C_{n' \dots} [n+n' \dots -1]^k (x+x' \dots)^{n+n' \dots -1-k}, \quad (13)$$

(where $r, r' \dots$ denote any positive or negative integer numbers satisfying $r+r'+\dots = k+1-p$, p being the number of terms in the series $n, n', \dots$). This equation constitutes a multinomial theorem of a class analogous to that of the exponential theorem contained in the equation (2).

In particular

$$C_{n+n' \dots} \Sigma_{r,r' \dots} [n-1]^r [n'-1]^{r'} \dots = C_n C_{n' \dots} [n+n' \dots -1]^k 2^{n+n' \dots -1-k}, \quad (14)$$

and if $p=2$, writing also m, n for n, n' , and $k-1-r$ for r' ,

$$C_{m+n} \Sigma_r [m-1]^r [n-1]^{k-1-r} = C_m C_n [m+n-1]^k 2^{m+n-1-k}, \quad (15)$$

or putting $k=0$ and dividing,

$$\frac{C_m C_n}{C_{m+n}} = \frac{1}{2^{m+n-1}} \Sigma_r [m-1]^r [n-1]^{-1-r}. \quad (16)$$

Now the series on the second side of this equation is easily seen to be convergent (at least for positive values of m, n). To determine its value write

$$F(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx;$$

then

$$F(m, n) = \int_0^{\frac{1}{2}} x^{m-1} (1-x)^{n-1} dx + \int_{\frac{1}{2}}^1 x^{m-1} (1-x)^{n-1} dx,$$

and by successive integrations by parts, the first of these integrals is reducible to

$$\frac{1}{2^{m+n-1}} \Sigma_r [m-1]^r [n-1]^{-1-r}, \quad r \text{ extending from } -1 \text{ to } -\infty \text{ inclusively,}$$

and the second to

$$\frac{1}{2^{m+n-1}} \Sigma_r [m-1]^r [n-1]^{-1-r}, \quad r \text{ extending from } 0 \text{ to } \infty;$$

hence

$$F(m, n) = \frac{1}{2^{m+n-1}} \Sigma_r [m-1]^r [n-1]^{-1-r},$$

or

$$C_m C_n \div C_{m+n} = F(m, n), \tag{17}$$

[Paper 102 continued in next slice.]